

1. For which values of x is the expression undefined?

$$\frac{3x-3}{x^2-49}$$

Any values that make the denom = 0

$$\begin{aligned} x^2-49 &= 0 \\ (x+7)(x-7) &= 0 \\ x+7=0 & \quad x-7=0 \\ \boxed{x=-7} & \quad \boxed{x=7} \end{aligned}$$

2. Find all vertical asymptotes of the following function.

$$f(x) = \frac{x^2-7x}{2x^2-6x-8}$$

Factors of the denom that do NOT cancel with factors of the numerator.

$$\begin{aligned} &= \frac{x(x-7)}{2(x^2-3x-4)} \quad \text{GCF!} \\ &= \frac{x(x-7)}{2(x-4)(x+1)} \\ &\quad \boxed{x=-4} \quad \boxed{x=-1} \end{aligned}$$

3. Find all holes of the following function. Write your answer as a coordinate point in simplest form. If no hole exists, write 'none'.

$$f(x) = \frac{x^2-16}{5x^2+29x+36} = \frac{x^2-16}{x^2+5x+4}$$

Factors of the denom that DO cancel with factors of the numerator

$$= \frac{(x+4)(x-4)}{(x+4)(x+1)}$$

Hole @ $x=-4$

$$f(x) = \frac{x-4}{x+1} \quad \text{Reduced Form!}$$

$$f(-4) = \frac{-4-4}{-4+1} = \frac{-8}{-3} = \frac{8}{3}$$

$$\boxed{\text{Hole @ } (-4, \frac{8}{3})}$$

4. Find all x -intercepts of the following function. Write your answer or answers as coordinate points.

$$f(x) = \frac{3x-12}{2x+3}$$

Values that make the numerator = zero BUT are not holes or V.A.

$$\frac{3(x-4)}{2x+3} \rightarrow 3(x-4)=0$$

$$x-4=0$$

$$x=4$$

$$\Rightarrow \boxed{(4,0)}$$

5. Find all y -intercepts of the following function. Express your answer as a coordinate point in simplest form.

$$f(x) = \frac{10x-16}{5x^2+32x-64}$$

The y value when $x=0$

$$f(0) = \frac{10(0)-16}{5(0)^2+32(0)-64}$$

$$= \frac{-16}{-64} = \frac{2}{8} = \frac{1}{4} \rightarrow \boxed{(0, \frac{1}{4})}$$

6. Find all horizontal asymptotes of the following function.

$$f(x) = \frac{3x^2-10x-48}{6x^2-57x+126}$$

"Even" degree

$$\rightarrow y = \frac{3}{6} = \frac{1}{2}$$

$$\boxed{y = \frac{1}{2}}$$

7. Find all horizontal asymptotes of the following function.

$$f(x) = \frac{x+2}{3x^2+27x+42}$$

"Bottom Heavy"

$$\rightarrow \boxed{y=0}$$

8. Find all horizontal asymptotes of the following function.

$$f(x) = \frac{x^2 + 7x - 30}{3x + 30}$$

"TOP Heavy"

→ None

9. Identify all vertical and horizontal asymptotes, x- and y-intercepts, and holes, and then use them to graph the rational function below.

$$f(x) = \frac{-3x + 6}{x^2 - 3x + 2} = \frac{-3(x-2)}{(x-2)(x-1)}$$

Hole @ $x=2$

$$f(2) = \frac{-3}{2-1} = -3$$

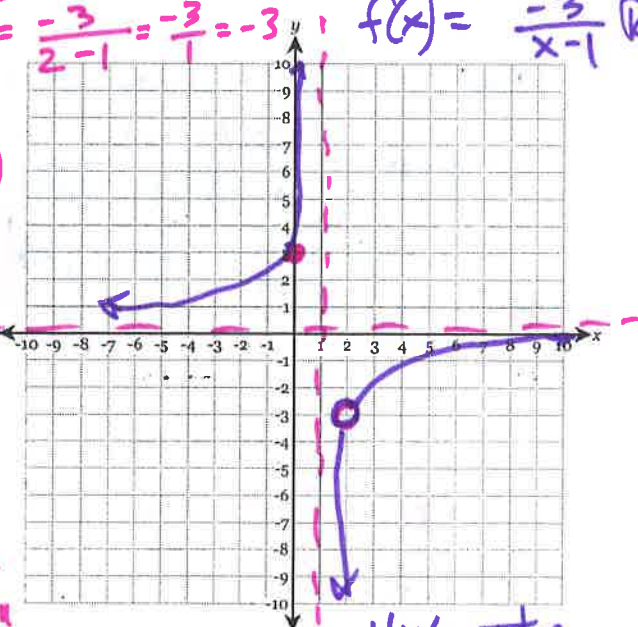
Hole

(2, -3)

$$y\text{-int} = \frac{6}{2} = 3$$

VA @ $x=1$

"Bottom Heavy"



→ HA @ $y=0$

HUG the asymptotes!

10. Simplify the expression completely if possible.

$$\frac{2x^3 - 8x^2}{6x^2}$$

Factor
Cancel

$$= \frac{2x^2(x-4)}{3 \cancel{6x^2}}$$

$$= \boxed{\frac{x-4}{3}}$$

11. Perform the operation and combine to one fraction.

$$\frac{3x+5}{x-1} + \frac{x}{x^2-1}$$

→ get common denom

$$\frac{(x+1)(3x+5)}{(x+1)(x-1)} + \frac{x}{(x+1)(x-1)}$$

$$= \frac{3x^2 + 5x + 3x + 5 + x}{(x+1)(x-1)}$$

$$= \boxed{\frac{3x^2 + 9x + 5}{(x+1)(x-1)}}$$

12. Perform the following operation and express in simplest form.

$$\frac{5x}{x+3} \div \frac{x^2 - 8x + 15}{x^2 - 9}$$

Flip and multiply

$$= \frac{5}{x+3} \cdot \frac{x^2 - 9}{x^2 - 8x + 15}$$

Factor

$$= \frac{5(x+3)(x-3)}{(x+3)(x-5)(x-3)}$$

$$= \boxed{\frac{5}{x-5}}$$